

```
# Loading the dataset using pandas

import pandas as pd

df = pd.read_csv('customer_shopping_behavior.csv')
```

```
df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                           3900 non-null   int64
1   Age                                    3900 non-null   int64
2   Gender                                3900 non-null   object
3   Item Purchased                        3900 non-null   object
4   Category                              3900 non-null   object
5   Purchase Amount (USD)                 3900 non-null   int64
6   Location                               3900 non-null   object
7   Size                                   3900 non-null   object
8   Color                                  3900 non-null   object
9   Season                                 3900 non-null   object
10  Review Rating                          3863 non-null   float64
11  Subscription Status                    3900 non-null   object
12  Shipping Type                          3900 non-null   object
13  Discount Applied                       3900 non-null   object
14  Promo Code Used                        3900 non-null   object
15  Previous Purchases                     3900 non-null   int64
16  Payment Method                         3900 non-null   object
17  Frequency of Purchases                 3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

```
# Summary statistics using .describe()
df.describe(include='all')
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000		3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN		
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN		N
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN		284
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065		Na
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983		Na
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000		Na
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000		Na
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000		Na
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000		Na
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000		Na

Checking if missing data or null values are present in the dataset

```
df.isnull().sum()
```

	0
Customer ID	0
Age	0
Gender	0
Item Purchased	0
Category	0
Purchase Amount (USD)	0
Location	0
Size	0
Color	0
Season	0
Review Rating	37
Subscription Status	0
Shipping Type	0
Discount Applied	0
Promo Code Used	0
Previous Purchases	0
Payment Method	0
Frequency of Purchases	0

dtype: int64

Imputing missing values in Review Rating column with the median rating of the product category

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

```
df.isnull().sum()
```

	0
Customer ID	0
Age	0
Gender	0
Item Purchased	0
Category	0
Purchase Amount (USD)	0
Location	0
Size	0
Color	0
Season	0
Review Rating	0
Subscription Status	0
Shipping Type	0
Discount Applied	0
Promo Code Used	0
Previous Purchases	0
Payment Method	0
Frequency of Purchases	0

dtype: int64

Renaming columns according to snake casing for better readability and documentation

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

```
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

```
# create a new column age_group
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels = labels)
```

```
df[['age', 'age_group']].head(10)
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

```
# create new column purchase_frequency_days
```

```
frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}
```

```
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

```
df[['purchase_frequency_days', 'frequency_of_purchases']].head(10)
```

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

```
df[['discount_applied', 'promo_code_used']].head(10)
```

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

```
(df['discount_applied'] == df['promo_code_used']).all()
```

```
np.True_
```

```
# Dropping promo code used column
```

```
df = df.drop('promo_code_used', axis=1)
```

```
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'previous_purchases', 'payment_method',
      'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],
      dtype='object')
```

✓ Connecting Python script to PostgreSQL

```
!pip install psycopg2-binary sqlalchemy
```

```
Requirement already satisfied: psycopg2-binary in /usr/local/lib/python3.12/dist-packages (2.9.11)
Requirement already satisfied: sqlalchemy in /usr/local/lib/python3.12/dist-packages (2.0.43)
Requirement already satisfied: greenlet>=1 in /usr/local/lib/python3.12/dist-packages (from sqlalchemy) (3.2.4)
Requirement already satisfied: typing-extensions>=4.6.0 in /usr/local/lib/python3.12/dist-packages (from sqlalchemy) (4.15.0)
```

```
from sqlalchemy import create_engine
```

```
# Step 1: Connect to PostgreSQL
```

```
# Replace placeholders with your actual details
```

```
username = "postgres"      # default user
```

```
password = "root" # the password you set during installation
```

```
host = "localhost"        # if running locally
```

```
port = "5432"             # default PostgreSQL port
```

```
database = "customer_behavior" # the database you created in pgAdmin
```

```
engine = create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}")
```

```
# Step 2: Load DataFrame into PostgreSQL
```

```
table_name = "customer"    # choose any table name
```

```
df.to_sql(table_name, engine, if_exists="replace", index=False)
```

```
print(f"Data successfully loaded into table '{table_name}' in database '{database}'.")
```

